Early Sepsis Recognition Based on Ear Localization using Infrared Thermography

Hasanain Al-Sadr
Electrical Engineering and Computer Science Department
University of Missouri Columbia
Computer Engineering
Columbia, MO, USA
Email: hoafkb@mail.missouri.edu

Mihail Popescu
Health Management and Informatics Department
University of Missouri Columbia
Computer Engineering
Columbia, MO, USA
Email: popescuM@health.missouri.edu

James M. Keller
Electrical Engineering and Computer Science Department
University of Missouri Columbia
Computer Engineering
Columbia, MO, USA
Email: kellerJ@missouri.edu

***Abstract*—Systematic screening is crucial for early diagnosis of sepsis. Detecting abnormal body temperature patterns can accurately predict sepsis before any other symptoms of infection. Therefore, we suggested using thermography as a non-invasive tool capable of continuously measuring body temperature patterns and detect abnormalities. One such pattern is the temperature difference (ETD) between body extremities such as outer ear or tip of the nose, and the core temperature such as the inside of the ear or the eye. Specifically, in sepsis, the ETD decreases from about 5-6F to about 1F. Thus, we propose a fully automated methodology for calculating the core vs. extremity temperature difference based on ear localization using thermal images and the fuzzy C-means (FCM) clustering algorithm. On a pilot dataset that consisted of 3674 face images from three human subjects, we achieved a 96.2% accuracy for ear localization. Consequently, we were able to accurately detect temperature differences between 1 and 6 F. The proposed methodology may be used in intensive care units (ICU) for early sepsis recognition.**

Keywords — Thermography, Sepsis, Ear Localization, FCM.

I. Introduction

Sepsis is defined as a clinical syndrome that consists of a dysregulated inflammatory response triggered by infection which has physiologic, pathologic and biochemical abnormalities [1]. Sepsis is the leading cause of patient mortality among non-coronary ICU patients and is the eleventh leading cause of death overall in the United States [2]. Sepsis represents a considerable healthcare burden, with a global rate of approximately 18 million cases per year and a mortality rate of 28–40% [3]. Each year, nearly $24 billion in healthcare costs are related with caring for patients with sepsis, making it the most expensive condition to treat in the U.S. healthcare system [3]. Data reported in 2016 showed that the annual costs for treating sepsis have increased $3.4 billion over the previous two-year period [4].

Prompt diagnosis of sepsis is essential. Initiating treatment of sepsis within the first six hours of presentation is critical for the optimization of patient outcomes [5]. The signs and symptoms of sepsis are non-specific, especially in the early phases of sepsis and are often missed due to multiple factors: the complex care required of ICU patients; attributing early signs to common post-operative problems; and lack of awareness of signs and symptoms [6]. If patients can progress to septic shock, the mortality rate is greater than 30%, despite aggressive interventions [6]. It is apparent that accurate screening of patients for sepsis is essential.

Sepsis detection methods are typically based on measuring a set of physiological variables such as thrombocytes, prothrombin time (PTT), sodium, potassium, lactate, etc. [7]. These methods are somewhat precise (area under the curve, AUC, around 90% [7]), they are not practical due to necessity of drawing blood too often. While temperature alone has shown poor sensitivity for identifying sepsis, its patterns have been found to be more promising [8]. Traditional methods of measuring body temperature only focus on temperature values in certain body points ignoring temperature distribution across body surface and time. In this paper we propose to use thermal imaging as a method to capture temperature trend and body distribution for sepsis detection.

Thermal imaging is a mobile, risk-free, pain-free, non-invasive tool capable of continuously measuring body temperature for the area of the body being imaged. Therefore, it can be used to detect different temperature distribution patterns in the region of interest (ROI). These patterns may show normal or some abnormality like pain, disease, inflammation [9]. Our hypothesis is that sepsis onset will lead to abnormal body temperature patterns which can be detected by calculating the difference between the core and the body extremities. Since the ICU patient's body is usually covered by a sheet and, maybe, an oxygen mask, we decided to compute ETD by evaluating the temperature difference between outer ear which represents core body temperature and the inside of the ear which represents a body extremity. In this paper we propose a robust ETD detection methodology based on a novel ear detection method and a novel FLIR One image processing method.

Our paper is organized as follows. In section II we present a literature review on medical thermography and ear detection methods. In section III we present our methodology. We

978-1-5386-5488-0/18/$31.00 ©2018 IEEE

present our results in Section IV and our conclusions in section V.

II. Literature Review

Patterns captured by using thermal camera can detect abnormality like pain, disease and inflammation [9]. In recent years, infrared thermography has been used as a clinical utility for breast cancer screening [7], [10]. In [11], a validation of thermography for studying vascular disorders was done using Doppler flowmetry after subject's hand is immersed in cold water. Similarly, Antonio-Rubio et al. [12] proposed a simple technique to evaluate hand autonomic dysfunction in Parkinson's disease (PD) by applying cold stress test. They found that PD patients reveal abnormal skin thermal responses at baseline and in response to cold. In [13], Singer et al proposed a method based on thermography to predict burn depth. This could facilitate the treatment decision for patients with burn injuries since superficial dermal and deep dermal burns have different treatment procedure. In [14], Hernandez-Contreras et al proposed a classification method to identify leg temperature patterns to distinguish between healthy and diabetic patients. De Salis et al. proposed a method to identify thoracic vertebral fractures in children with osteogenesis imperfecta based on high resolution thermal imaging [15].

In [16], Kanazawa et al. investigated the validity and reliability of using FLIR One camera first generation for diabetic foot ulcer assessment by comparing it with a standard live clinical assessment. Hardwicke et al. [17] evaluate blood flow in patients undergoing free flap reconstruction. Then, a suitable perforator is selected based on the rate and rewarming pattern. In [18], thermal based method is proposed to overcome the technical challenges involving the reconstruction of the lower limb after skin cancer surgery. In [19], a comparison between Smartphone-based thermal camera and conventional infrared probe temperature scanner has been conducted for tissue temperature measurement in hand and upper extremity surgeries.

Ear detection methods can be divided into four different categories. The first category includes ear detection models that are based on active contour idea. Burge and Burger [20] detect ear by applying deformable contours on the image gradient and used thermal images to detect occlusions since subject's hair has lower temperature than the ear. In [21], Alvarez et al. proposed an ear detection method based on the combination of geodesic active contours model and ovoid model, and Deepak et al. [22] address the issue of varying background and face poses by using a snakes as active contour model.

The second detection category is based on the elliptical shape of the ear. In [23], Canny edge detector and reduced Hough transform are used to create an accumulator with multiple peaks. Then, the ear is detected by removing the peaks that are corresponding to non-vertical ellipses. Cummings et al. [24] enhance image circular features by applying image ray transform. The ear is localized by applying smoothing, thresholding and elliptical template matching. In [25], the tip of the nose is used to estimate the ear region. The ear is localized by applying oval template matching. Yuan and Mu [26] applied skin- color detector and intensity contour formation to estimate ear location. The ear is detected by fitting an ellipse to the image edges.

The third localization category is based on template matching model. Sana et al. [27] proposed a template-based ear detection technique that uses different template sizes to localize ears at different scales. Prakash and Gupta [28] proposed an automatic ear localization technique that is invariant to scale, rotation and shape where ear is localized by comparing multiple ear candidates generated by connected components with the created off-line ear template. Halawani [29] proposed a template-based ear localization technique that is based on a single deformable contour template.

The fourth detection category uses discriminative classification models. Islam et al. [30], modified the cascaded AdaBoost face detection (Viola-Jones) to detect ear from side face images. This is done by generating weak classifiers based on Haar-like rectangular features. Then final detector was built by cascading classifiers that accept a sub-window of the ear image. Yuan and Mu [31] improved the AdaBoost algorithm by introducing structural features of the ear. Abaza and Bourlai in [32] proposed a thermal ear detection approach based on cascaded AdaBoost to overcome the illumination problem. In [33], a brief review of deep learning-based (CNN) ear recognition techniques are presented. Also, a technique to enhance the speed of the CNN is proposed by optimizing the sliding window approach for ear detection. In [34], a CNN ear detection model is developed where three network architectures are investigated.

Based on the method categories listed above, some specific notes and limitations associated with each one are worth mentioning:

- Active contour model cannot be used in fully automated ear localization since manual contour initialization is required.
- In elliptical ear model, the assumption of elliptical ear shape is not correct since some of the ears have different shapes (round, oval, triangular and rectangular) [35]. Also, introducing different face poses with different angles will result in irregular ear shape on which elliptical shape model may fail.
- For template matching model, there is no template set that can handle all real-world situations.
- In classification model, a large training dataset is required as well as a pre-training model in Deep Learning approaches is required too.
- Additionally, some of these techniques are not fast enough for real-time applications and are applied to image in a control environment with minimum background and limited face poses.

To overcome the limitations listed above, we propose a fully automated unsupervised model for ear localization based on using thermal image and Fuzzy C-means (FCM) clustering algorithm.

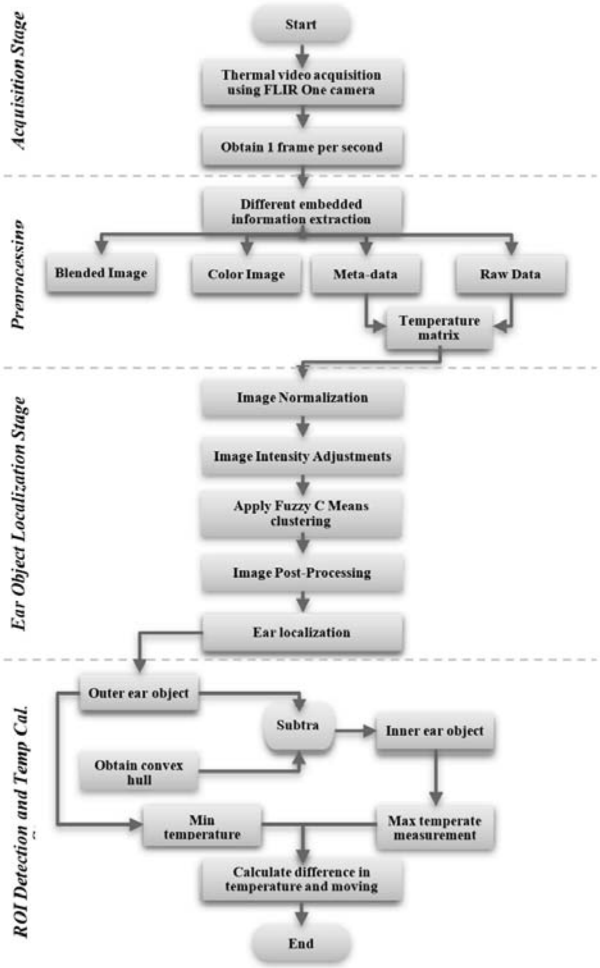

Fig. 1. The proposed ETD detection method.

III. Methodology

In this study, we propose a robust method for detecting temperature patterns differences between core (inner ear) and body extremities (outer ear) using low cost thermal imaging. We hypothesize that the measurement of the temperature difference across the ear (ETD) can be used for predicting sepsis. In Fig. 1 we provide a summary diagram of our ETD detection methodology. The proposed methodology consists of four main stages: data acquisition stage, preprocessing stage, ear object localization stage, ROI detection and ETD calculation stage.

A. Data Acquisition and Preprocessing Stage

Thermal images are obtained from a FLIR One camera[1] that is capable of capture both thermal and color images. It uses a Lepton thermal sensor which is a longwave infrared (LWIR) sensor with a resolution of 160×120 pixels. The acquired images are stored using multi-spectral dynamic (MSX) blending technology that embeds the visible image (640×480 resolution) with the thermal image after digitally scaled (320×240 resolution). The thermal camera can capture a scene temperature range between -4 to 248 F and can detect changes as small as 0.18 F. It also has a mechanical shutter that performs automatic tuning periodically which keeps the thermal camera calibrated. The thermal camera can stream up to 9 frames per second (fps). However, we found that it is sufficient to use 1 frame per second without affecting the temperature patters response. Also, this will result in faster processing and less storage space required.

Working only with the MSX blending images that were produced by FLIR One camera have some limitations. First, the MSX technology suffered from misalignment between the visible and the thermal image. This is due to the difference in size and field of view between the thermal and the visible image. Second, representing the additional range of the electromagnetic spectrum as pseudo colormap reveal the radiation differences in visible spectrum but it cannot be accurately used to quantify the change in temperature distribution over time. The pseudo color mapping depends directly on temperature range and the color scale of the scene which known as color palette. The palette maps the pixels based on their relative values instead of their absolute temperature value. Consequently, introducing a new object in the scene with higher/lower than the scene temperature range will affect the temperature color distribution. For the same thermal image, different color visualizations can be generated by using different color palettes. It is also possible to use fixed palette where certain temperatures ranges assigned to specific colors. Regardless which color palettes are used, some information will be lost due to the quantization process and it cannot be recovered. Third, FLIR One stores a modified version of the Lepton thermal sensor reading by using a nonlinear scale factor of two to provide a crisper thermal image. However, any scaling or modification in the actual sensor will have negative impact on the accuracy of the reading. This is especially important in our case since the number of the pixels in the targeted ROI are small. As a result, to get accurate reading from this camera, the temperature matrix should be extracted based on the actual raw IR sensor data. Working directly with the thermal image matrix will give the ability to detect and quantify changes in temperature distribution over time for the target ROI.

Thermal camera measures the infrared radiation emitted by an object by interpreting the received energy into an electronic voltage. The received signal contains not only radiation form the target object but also from the surrounding objects and the atmosphere. Therefore, the voltage generated by the camera detector (s(T)), is assumed to be a function of three quantities: the desired radiation emitted by the object (E_{obj}), the ambient radiation reflected in the object (E_{ref}) and the emission of the atmosphere between the object and the camera (E_{atm}) [36]. This process is illustrated in Fig. 2 and expressed in 1.

[1] https://www.flir.com/

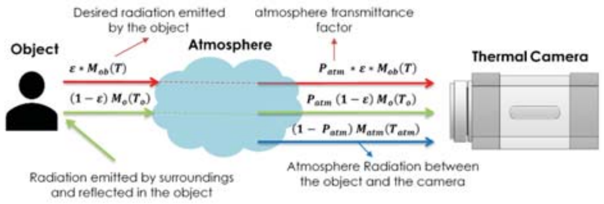

Fig. 2. Total radiation received by the thermal camera.

$$s(T) = K_{(DetectorSensitivity)}(E_{obj} + E_{ref} + E_{atm}) \quad (1)$$

The main principle of thermal camera operation is based on the amount of radiation energy emitted by a blackbody at temperature T per unit time, surface area, and wavelength interval. This is also known as Planck's law of black body radiation. The total blackbody emissive power can be calculated by integrating the spectral blackbody emissive power over the operating wave range of the camera detector as shown in (2-7) [37].

$$M_{bb}(T) = \int_{\lambda_1}^{\lambda_2} \frac{c_1}{\lambda^5[\exp\frac{c_2}{\lambda T} - 1]} d\lambda \quad (2)$$

$$C_1 = 2\pi hc^2 = 3.74177 \times 10^8 \ \ W.\mu m^4/m^2$$
$$C_2 = hc/k = 1.43878 \times 10^4 \ \ \mu m.K,$$

where $c = 3 \times 10^8 m/s$ is the speed of light, $h = 6.626 \times 10^{(}-34)J/s$ is the Planck constant, $k = 1.38 \times 10^{(}-23)J/K$ is Boltzmann constant and C_1 and C_2 are the first and second radiation constant respectively. Real objects radiate less energy than the blackbody when operate in the same temperature. The ratio of the real body radiation emitted to radiation emitted by a blackbody is called emissivity ϵ.

$$M(T) = \epsilon \times M_{bb}(T) \quad (3)$$

The signal generated by the camera (raw data pixels) is assumed to be a function of three quantities:

$$\begin{aligned} s(T) = K.[\epsilon \times P_{atm} \times M_{ob}(T) + (1-\epsilon)P_{atm} \times M_o(T_o) \\ +(1-P_{atm})M_{atm}(T_{atm})] \end{aligned} \quad (4)$$

Where ϵ is the object emissivity (for human skin ϵ =0.98), P_{atm} is the atmosphere transmittance factor, M_{ob} is the object emittance, M_o is the ambient exitance and M_{atm} is the atmospheric emittance. The radiation emitted by the object can be calculated:

$$\begin{aligned} M_{ob}(T) = \frac{s(T)}{K \times \epsilon \times P_{atm}} - \frac{(1-\epsilon) \times M_o(T_o)}{\epsilon} \\ - \frac{1 - P_{atm} M_{atm}(T_{atm})}{\epsilon \times P_{atm}} \end{aligned} \quad (5)$$

Also, the ambient and atmospheric emittance can be estimated by using RBF:

$$M_o(T_o) = \frac{R}{\exp\frac{B}{T_o} - F} \quad (6)$$

$$M_{atm}(T_{atm}) = \frac{R}{\exp\frac{B}{T_{atm}} - F} \quad (7)$$

The constants R, B, F are specific for each camera based on the type of optical equipment and measuring range. They are defined in the calibration process. The transmission of the atmosphere (P_{atm}) is a function of the relative humidity (ω %), atmospheric temperature (T_{atm}) and the distance from the camera to the object (d) and can be define in (8-10) [38].

$$\begin{aligned} P_{atm} = \frac{X}{\exp[\sqrt{d}(\alpha_1 + \beta_1 \times \sqrt{Cof_{H_2O}}]} + \\ \frac{1-X}{\exp[\sqrt{d}(\alpha_2 + \beta_2 \times \sqrt{Cof_{H_2O}}]} \end{aligned} \quad (8)$$

The constants X, α, β are determined at the stage of camera calibration for a specific temperature range (FLIR One -20 to 120 Celsius). Air temperature (T_{atm}) and the relative humidity are needed to calculate the Cof_H_2O.

$$\begin{aligned} COf_{H_2O} &= \omega\% \times e^{(A \times B \times C \times D)} \\ A &= 1.5587 + 6.939 \times 10^{(-2)} \\ B &= T_{atm} - 2.7816 \times 10^{(-4)} \\ C &= T_{atm}^2 + 6.8455 \\ D &= 10^{(-7)} \times T_{atm}^3 \end{aligned} \quad (9)$$

Finally, the target object temperature is obtained from the transformation of the RBF model:

$$T_{ob} = \frac{B}{ln(\frac{R}{M_{ob}} + F)} \quad (10)$$

As a result, the object temperature matrix can be calculated by specifying the: object emissivity, ambient temperature, atmospheric temperature, humidity and the distance from the camera to the object. Through the experiments, the measurement of relative humidity was around 52% and the distance from the camera to the ear is set at 1.2 m. The emissivity level was fixed at 0.98, which is the emissivity of human skin. Finally, the atmospheric temperature was around 24.5 degree Celsius.

B. Ear Object Localization Stage

1) Preprocessing Step: In this step, the thermal image matrix is normalized in [0,1] as shown in Eq. 11. Then, the contrast level of the normalized image is enhanced by saturating 1% of the top and bottom pixels. The result of image adjustment is show in Fig. 3.

Fig. 3. Total radiation received by the thermal camera.

$$NormalizedImage = \frac{Raw - Min(Raw(:))}{Max(Raw(:)) - Min(Raw(:))} \tag{11}$$

2) Ear Localization Step: The localization of human ear in the thermal image is done in segmentation process by using FCM clustering. Our proposed ear localization approach is based on the fact that a substantial temperature difference exists between the ear and the rest of the head. Three different clustering algorithms have been tested: K means, FCM, and fuzzy local information C-Means (FLICM). However, FCM gave us the best results for overlapping regions comparing with the other two algorithms, hence we used it in this paper as a first step for ear localization. In FCM image segmentation, the image is grouped into a specific number of clusters, C, where every pixel in the image belongs to each cluster to some degree based on the membership value [39]. FCM membership $u \in 0, 1$ means that a pixel can have partial membership to multiple clusters. The number of clusters depends C on the type of the problem, type of the analysis and the expected use of the results. Here, after analyzing several images, we selected C=3, as shown in Fig. 4. To obtain stable and repeatable clustering results, the clusters are sorted based on their temperature values. Since the scene background has the lowest temperature values, it will always be in cluster one. Accordingly, pixels corresponding to exposed skin will have the highest temperature values and will be placed in cluster three. Finally, the remaining pixels including pixels of subject's ear will lies in cluster two.

Our ear localization approach is based on the fact that there will always be a significant temperature difference between the pixels of the subject's ear and the surrounding pixels of the head. It can be seen from Fig. 4c, where the ear is part of cluster two. However, it may be difficult to find a way to localize the ear from cluster two that can adapt to all images especially in an uncontrolled environment. This is because there is a small difference in temperature distributions between regions within the same cluster. Therefore, cluster three will be used for ear localization considering that subject's head where ear is located will always be part of cluster three. We suggest a simple way to extract ear efficiently from cluster three if it is surrounding by white pixels. However, in most cases, the ear is not fully surrounded by white pixels because pixels that represent hair may be in different cluster. The

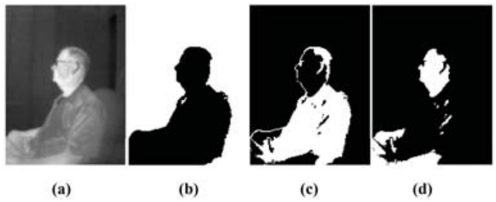

Fig. 4. FCM clustering results (a) Adjusted image (b) Cluster 1 (c) Cluster 2 (d) Cluster 3

ear preprocessing steps are shown in Fig. 5. First, the image is thresholded based on the normal body core temperature ranges and the region with the highest value that represents subject's head is detected as shown in Fig. 5a. Then, closing operation with disk structure element of size 7 is used to fill up the ear region while preserving the geometrical features as shown in Fig. 5b. After that, the boundary contour of the face is extracted from binary image based on 8 connected neighborhood pixels as shown in Fig. 5c. Finally, the boundary mask is superimposed to the original image to create a new image where ear is fully surroundings by white pixels as shown in Fig 5d.

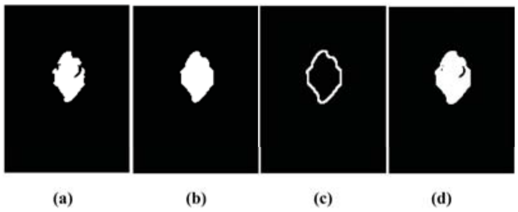

Fig. 5. Preprocessing steps for ear localization (a) Face detection based on cluster 3 (b) After applying closing operation (c) Face boundary detection (d) Superimposed image.

3) ROI Detection and Temperature Calculation: The complement of superimposed image is calculated to obtain a white ear on a dark face with white background as shown in Fig 6b. Next, the localization of human ear is done in by using connected components labeling where it scans the image and groups its pixels into components based on pixel connectivity. Then, a sorted list of strongly connected components is generated. As background has the largest area component in the image, the ear will be the second larger area and the location information is extracted as shown in Fig 6c.

After ear is localized, the outer ear is automatically cropped based on the detected bounding box information. Next, the smallest set of the points that define the ear boundary is calculated using convex hull algorithm as shown in Fig 7a. Then based on the contour points, a binary convex hull mask is generated as shown in Fig. 7b. The inner ear area is detected by subtracting the outer ear image from the convex hull mask that produced in the previous step as shown in Fig 7c.

Temperature difference (ETD) between outer and the inner ear have been computed based on the two ear masks that were generated previously. Finally, core body temperature is calculated by taking the maximum temperature of inner ear part. Similarly, minimum temperature of the outer ear part is computed which represent temperatures in the body's extremity. Minimum and maximum pixels locations are shown in Fig. 8.

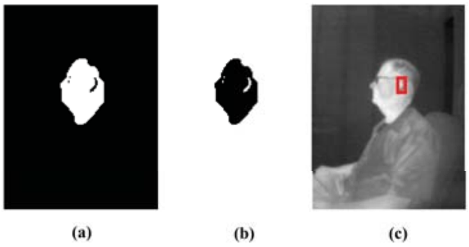

Fig. 6. Ear localization steps: (a) Preprocessed image (b) Complement image (c) Ear localization.

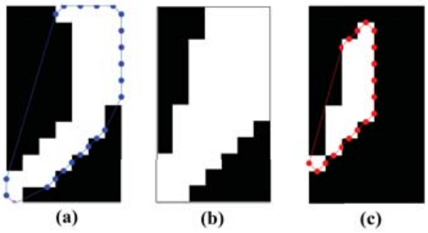

Fig. 7. Outer and inner part of outer ear detection (a) Outer ear part (b) convex hull image (c) Inner ear part.

Fig. 8. Ear localization steps: (a) Preprocessed image (b) Complement image (c) Ear localization.

IV. Experimental Setup and Results

A. Experimental Results

All testing was completed in the same location using the same equipment with room temperature maintained around 24.5 Celsius. Each subject (two of the investigators) was asked to sit comfortably on a chair and the camera was mounted on a tripod about 1.2 m from the subject side face. To maintain a stable upper-body position, the subject was instructed to put his hands in the two empty containers that were placed on the left and the right side of the subject. Pre-immersion calculation was done by accruing thermal and visible images of the subject at 1 frame per second for 3 min to identify a baseline. Then the two containers were replaced with other containers of 12 Celsius water and the subject immersed his hands for 3 min. After 3 min of immersion in ice water, the subject removed his hands and placed it again on the empty containers with towel to dry the subject hand and the camera acquired images at 1 fps for the next 20 min. The protocol was repeated for each of the research group members and the acquired data are stored into dataset with different image formats.

B. Thermal Image Segmentation

The goal of the ear finding stage is to localize the ear that can be further processed for inner and outer ear part detection. Since the temperature pattern is calculated based on the differences between core (inner ear) and body extremities (outer ear) and we want to automatically track changes in the temperature pattern. Furthermore, the tracking is done in an unconstrained environment which means that we are expected to have errors due to frames where the ear covered or not in the scene, face positioned at some angels where no inner part can be detected, or false ear detections. Therefore, it is crucial to extract correct masks for both inner and outer ear part and eliminate any frames that will not serve our purpose. We tested our ear localization system using 3674 unconstrained face profile image sequences for two different cases. The proposed system achieved accuracy of 96.2% correct ear localization. A sample image of the detected inner and outer ear is shown in Fig. 9.

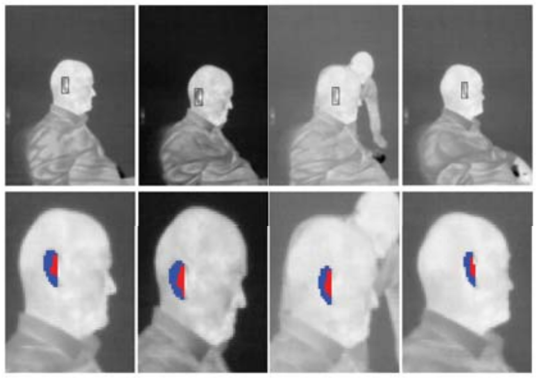

Fig. 9. Examples of correct ear localization results obtained based on the proposed system.

Any false detected ear will have high impact on the temperature pattern. Therefore, three constrain are enforced to eliminate all the failed detection automatically. First, we constrain the temperature of the detected ear within the body temperature range (between 27 to 39 Celsius). Second, a spatial constraint is enforced by tracking the overlap area of the detected bounding box in the consecutive frames. If there is no overlap between two consecutive frames, the frame will be eliminated from the temperature pattern calculations. On the other hand, some of the positive detected ear may be eliminated if the subject adjusts his/her position quickly. Finally, a size constraint is implemented by making sure that the difference between the area of the bounding box with respect to the average detected bounding box is no grater that 50%. Example of eliminated frames based on the above constrains are illustrated in Fig. 10.

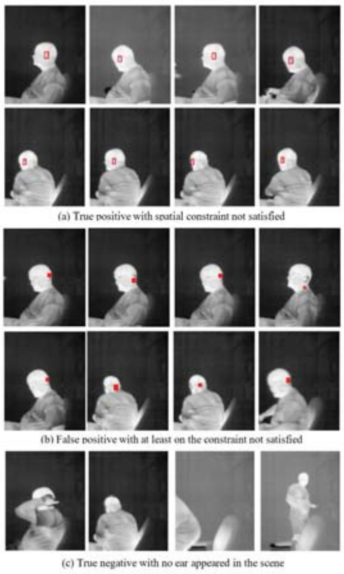

Fig. 10. Examples of eliminated frames where one or more constraints are not satisfied.

C. Ear Temperature Distribution Patterns

First, we need to address the question: is there any evidence that ETD in sepsis becomes small? In a study (unpublished) performed at the University of Missouri Hospital with IRB approval, we collected IR data from 11 ICU patients. Of the 11 patients followed, only one ended up becoming septic. In Fig. 11 we show the ETD for this patient.

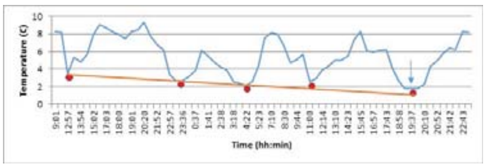

Fig. 11. Temperature difference (ETD, blue) and its minima (orange), 36 hours before sepsis (marked by arrow).

From Fig. 11 we see that ETD minimums (red points) decreased continuously 36 h before sepsis started (arrow). The patient was treated with antibiotics and recovered (ETD not shown).

As a proxy for sepsis, in this study we use ”cold stress” induced by Cold Water Immersion (CWI). The effect of the ”cold stress” will result in an abnormal ear temperature patterns that can be used to simulate body response to sepsis. Therefore, the minimum and maximum temperature difference between the outer and inner ear part are calculated and plotted with respect to time throughout the experimental duration as shown in Fig. 12. The moving average is calculated over two minutes period to identify the primary temperature trend. The first three minutes represents the baseline data where the subject sit comfortably without immersing his hand in cold water. It is important to detect the baseline since the process of monitoring changes in temperature is done with respect to baseline. Fig.11a shows a stable trend with small variations in the baseline collecting stage. The results also show that different cases have different baseline. After the subject immerse his/her both hands, the temperature difference increases gradually. Even after applying the CWI, the temperature keeps increasing until reaches a maximum peak which is approximately 1.2 Celsius degree from the baseline for subject one. Finally, the recovery stage is started where temperature difference start to decrease until return to baseline values.

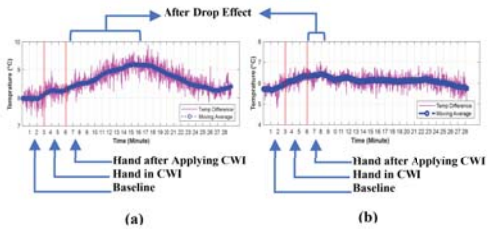

Fig. 12. The temperature difference between the core and the body extremities (a) Subject 1 (b) Subject 2.

The results of this study are in line with previous studies that explore the body response to CWI. Generally, CWI will result in high rate of heat transfer from skin to water which will translate to temperature drops in the extremities while keeping the core body temperature fixed [40]. However, if the subject is exposed to CWI for long time, core temperature begins to decline as well. Even though we observed that the core temperature also decreased when extremities temperature decreases but with much lower rate as shown in Fig. 13. The body response to CWI by keeping the main core organs warm and limiting the blood flow to the extremities. The rate of core temperature variation is corresponding to the balance between heat loss to the water and heat produce by the body. After applying CWI, the warm blood from the body core is circulate back to the colder extremities which will result in after drop effect where body temperature continues to decrease (difference increase) until reach a specific point. After this point, the recovery stage will start where temperature increases (difference decrease) until reach the baseline level. To show the temperature trend stated above, the average temperature of the outer and inner ear region is calculated and plotted for subject 1 as illustrated in Fig 13.

V. Conclusions

In this paper, we presented a methodology for detecting the temperature differential (ETD) across human ears that can be used for early sepsis detection. Toward this goal, we proposed a fully automated unsupervised model for inner and outer ear localization based on low cost infrared thermography. The results demonstrated the robustness of the technique in localizing inner and outer ear part and eliminating false detection which can successfully handle various poses and varying background and can be used in real-time application. The result of temperature pattern calculation is in line with previous studies that explore the body response to CWI and indicate that thermal imaging could be an effective tool in medical diagnosis.

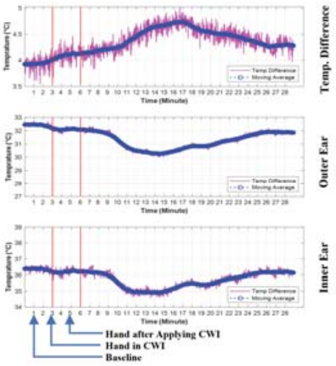

Fig. 13. The average temperature of the outer and inner ear region for subject one.

REFERENCES

[1] R. Neviere, "Sepsis syndromes in adults: Epidemiology, definitions, clinical presentation, diagnosis, and prognosis," *Monografía en Internet]. Wolters Kluwer: UpToDate*, 2017.

[2] J. Yan, S. Li, and S. Li, "The role of the liver in sepsis," *International reviews of immunology*, vol. 33, no. 6, pp. 498–510, 2014.

[3] S. Cantrell, "Technology takes the lead in reducing sepsis," *Infection*, 2018.

[4] G. S. Martin, "Sepsis, severe sepsis and septic shock: changes in incidence, pathogens and outcomes," *Expert review of anti-infective therapy*, vol. 10, no. 6, pp. 701–706, 2012.

[5] R. L. Gauer, "Early recognition and management of sepsis in adults: the first six hours." *American family physician*, vol. 88, no. 1, 2013.

[6] L. J. Moore, K. L. Turner, and S. R. Todd, *Common problems in acute care surgery*. Springer, 2013.

[7] S. Pramanik, D. Bhattacharjee, and M. Nasipuri, "Multi-resolution analysis to differentiate the healthy and unhealthy breast using breast thermogram," in *Systems in Medicine and Biology (ICSMB), 2016 International Conference on*. IEEE, 2016, pp. 49–52.

[8] A. M. Drewry, B. M. Fuller, T. C. Bailey, and R. S. Hotchkiss, "Body temperature patterns as a predictor of hospital-acquired sepsis in afebrile adult intensive care unit patients: a case-control study," *Critical Care*, vol. 17, no. 5, p. R200, 2013.

[9] S. Bardhan, M. K. Bhowmik, S. Nath, and D. Bhattacharjee, "A review on inflammatory pain detection in human body through infrared image analysis," in *Advanced Computing and Communication (ISACC), 2015 International Symposium on*. IEEE, 2015, pp. 251–257.

[10] H. G. Zadeh, J. Haddadnia, N. Ahmadinejad, and M. R. Baghdadi, "Assessing the potential of thermal imaging in recognition of breast cancer," *Asian Pac J Cancer Prev*, vol. 16, pp. 8619–8623, 2015.

[11] L. Stefańczyk, B. Woźniakowski, P. Pietrzak, A. Majos, and P. Grzelak, "Comparison of thermography and doppler sonography in the evaluation of the cold immersion test in women with excessive vasospastic reaction," *Medical Science Monitor*, vol. 13, no. 1, pp. 121–128, 2007.

[12] I. Antonio-Rubio, C. Madrid-Navarro, E. Salazar-López, M. Pérez-Navarro, C. Sáez-Zea, E. Gómez-Milán, A. Mínguez-Castellanos, and F. Escamilla-Sevilla, "Abnormal thermography in parkinson's disease," *Parkinsonism & related disorders*, vol. 21, no. 8, pp. 852–857, 2015.

[13] A. J. Singer, P. Relan, L. Beto, L. Jones-Koliski, S. Sandoval, and R. A. Clark, "Infrared thermal imaging has the potential to reduce unnecessary surgery and delays to necessary surgery in burn patients," *Journal of Burn Care & Research*, vol. 37, no. 6, pp. 350–355, 2016.

[14] D. Hernandez-Contreras, H. Peregrina-Barreto, J. Rangel-Magdaleno, and J. Gonzalez-Bernal, "Narrative review: Diabetic foot and infrared thermography," *Infrared Physics & Technology*, vol. 78, pp. 105–117, 2016.

[15] A. F. De Salis, R. Saatchi, and P. Dimitri, "Evaluation of high resolution thermal imaging to determine the effect of vertebral fractures on associated skin surface temperature in children with osteogenesis imperfecta," *Medical & biological engineering & computing*, pp. 1–11, 2018.

[16] T. Kanazawa, G. Nakagami, T. Goto, H. Noguchi, M. Oe, T. Miyagaki, A. Hayashi, S. Sasaki, and H. Sanada, "Use of smartphone attached mobile thermography assessing subclinical inflammation: a pilot study," *Journal of wound care*, vol. 25, no. 4, pp. 177–182, 2016.

[17] J. T. Hardwicke, O. Osmani, and J. M. Skillman, "Detection of perforators using smartphone thermal imaging," *Plastic and reconstructive surgery*, vol. 137, no. 1, pp. 39–41, 2016.

[18] S. P. Paul, "Using a thermal imaging camera to locate perforators on the lower limb," *Archives of plastic surgery*, vol. 44, no. 3, p. 243, 2017.

[19] J. Cao, K. Currie, P. Carry, G. Maddox, S. Nino, and K. Ipaktchi, "Smartphone-based thermal imaging: A new modality for tissue temperature measurement in hand and upper extremity surgeries," *HAND*, vol. 13, no. 3, pp. 350–354, 2018.

[20] M. Burge and W. Burger, "Ear biometrics," in *Biometrics*. Springer, 1996, pp. 273–285.

[21] L. Alvarez, E. González, and L. Mazorra, "Fitting ear contour using an ovoid model," in *Security Technology, 2005. CCST'05. 39th Annual 2005 International Carnahan Conference on*. IEEE, 2005, pp. 145–148.

[22] R. Deepak, A. V. Nayak, and K. Manikantan, "Ear detection using active contour model," in *Emerging Trends in Engineering, Technology and Science (ICETETS), International Conference on*. IEEE, 2016, pp. 1–7.

[23] B. Arbab-Zavar and M. S. Nixon, "On shape-mediated enrolment in ear biometrics," in *International Symposium on Visual Computing*. Springer, 2007, pp. 549–558.

[24] A. H. Cummings, M. S. Nixon, and J. N. Carter, "A novel ray analogy for enrolment of ear biometrics," in *Biometrics: Theory Applications and Systems (BTAS), 2010 Fourth IEEE International Conference on*. IEEE, 2010, pp. 1–6.

[25] K. V. Joshi, "Oval shape detection and support vector machine based approach for human ear detection from 2d profile face image," *International Journal of Hybrid Information Technology*, vol. 7, no. 5, pp. 113–120, 2014.

[26] Z.-C. Mu *et al.*, "Ear detection based on skin-color and contour information," in *Machine Learning and Cybernetics, 2007 International Conference on*, vol. 4. IEEE, 2007, pp. 2213–2217.

[27] A. Sana, P. Gupta, and R. Purkait, "Ear biometrics: A new approach," in *Advances in Pattern Recognition*. World Scientific, 2007, pp. 46–50.

[28] S. Prakash and P. Gupta, "An efficient ear localization technique," *Image and Vision Computing*, vol. 30, no. 1, pp. 38–50, 2012.

[29] A. Halawani and H. Li, "Human ear localization: A template-based approach," in *Proceedings of the International Workshop on Pattern Recognition (ICOPR 2015), Dubai, UAE*, 2015, pp. 4–5.

[30] S. M. Islam, M. Bennamoun, and R. Davies, "Fast and fully automatic ear detection using cascaded adaboost," in *Applications of Computer Vision, 2008. WACV 2008. IEEE Workshop on*. IEEE, 2008, pp. 1–6.

[31] L. Yuan and Z. Mu, "Ear recognition based on gabor features and kfda," *The Scientific World Journal*, vol. 2014, 2014.

[32] A. Abaza and T. Bourlai, "On ear-based human identification in the mid-wave infrared spectrum," *Image and Vision Computing*, vol. 31, no. 9, pp. 640–648, 2013.

[33] P. L. Galdámez, W. Raveane, and A. G. Arrieta, "A brief review of the ear recognition process using deep neural networks," *Journal of Applied Logic*, vol. 24, pp. 62–70, 2017.

[34] Ž. Emeršič, D. Štepec, V. Štruc, and P. Peer, "Training convolutional neural networks with limited training data for ear recognition in the wild," *arXiv preprint arXiv:1711.09952*, 2017.

[35] S. Prakash and P. Gupta, *Ear biometrics in 2D and 3D: localization and recognition*. Springer, 2015, vol. 10.

[36] L. Michalski, K. Eckersdorf, J. Kucharski, and J. McGhee, *Temperature measurement*. John Wiley & Sons, 2001.

[37] S. GRYŚ, "Image and data processing techniques in characterization of material nonuniformity by active ir thermography," 2013.

[38] "Toolkit ic2 dig16 developers guide 1.01 for agema 550/570," *Thermo-Vision Family FLIT System*, 2001.

[39] A. N. Kataria, D. M. Adhyaru, A. K. Sharma, and T. H. Zaveri, "A survey of automated biometric authentication techniques," in *Engineering (NUiCONE), 2013 Nirma University International Conference on*. IEEE, 2013, pp. 1–6.

[40] D. Clauss. (2011) The physiologic effects of cold water immersion. [Online]. Available: http://lakeice.squarespace.com/the-medical-view/